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SHORT COMMUNICATION

A larval transcriptome of the limnephilid caddisfly *Micropterna lateralis* (Stephens, 1837) (Trichoptera: Limnephilidae)

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ABSTRACT

The aquatic caddisfly *Micropterna lateralis* (Stephens, 1837) is adapted to intermittent streams. Normally, the species persists drought conditions by means of an adult diapause, yet active larvae, pupation and emergence have been observed on dry stream beds. The species, thus, presents a good example to study the molecular basis of local adaptation to different hydrological conditions and freshwater habitat types, but genomic resources are lacking to date. We isolated total RNA and applied Illumina RNA-Seq to a pool of eight larval specimens sampled under different hydrological conditions during a drought experiment simulating stream intermittency. Sequencing resulted in 108'011'306 high-quality reads that were *de novo* assembled into 17,759 contigs, of which 16,848 were matched to protein sequences from Insects. The transcriptome of *M. lateralis* is providing a valuable resource for future investigations of molecular mechanisms in drought adaptation in aquatic insects and the secondary invasion of freshwater habitats by insects in general.

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Trichoptera; Limnephilidae;
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Introduction

Caddisflies are one of the most diverse insect orders displaying high levels of ecological diversity amongst and occasionally within species (Mackay and Wiggins 1979). The widespread, case-bearing limnephilid *Micropterna lateralis* (Stephens, 1837) is abundant in European freshwaters, inhabiting both permanently flowing and intermittent streams (Graf, Murphy, Dahl, Zamora-Munoz, and López-Rodríguez 2008). The occurrence in differing habitats makes *M. lateralis* an interesting model to study the level of plasticity and local adaptation to hydrological conditions. Genomic and transcriptomic data could be used to illuminate the molecular mechanisms underlying this plasticity or adaptation (e.g., Stapley et al. 2010), but such resources are scarce for Trichoptera (Misof et al. 2014; Peters et al. 2014). Here, we present the larval transcriptome of a limnephilid caddisfly as an important starting point for studies targeting comparative biology and differential gene activity under various environmental conditions and during different developmental stages.

Biological samples

Micropterna lateralis larvae were collected in the Hessian Spessart (Auerbach, 50°16'34.2"N, 9°25'58.7"E) in November 2014, and transported to the laboratory on ice within two hours. They were allowed to acclimate for three days in aerated aquaria (3000-ml glass crystallising dishes with two airstones, 900-mL sediment, 700-mL water and 3-g naturally conditioned leaf litter; all collected at the study stream) that were kept in climate chambers with a day/night cycle of L10:D14, temperature set to 14°C and humidity kept at 80% prior to common garden experiments. During the subsequent experiment, humidity was reduced to 30% to simulate gradual drying through evaporation. Larvae were held in groups of five within individual aquaria. The experiment was run for seven days with samples being taken at irregular intervals (day in experiment; time of day) to cover various hydrological conditions (from wet to dry) and daytimes to maximise the diversity of transcripts (Figure 1). Sampled caddisfly larvae were shock frozen in liquid nitrogen and stored at −80°C prior to RNA extraction.

Isolation of RNA and Illumina sequencing

Entire frozen larvae were used for the isolation of total RNA with TRIzol Reagent (Thermo Fisher Scientific) and an RNeasy Mini Kit (Qiagen, Hilden, Germany) following the protocol of Ann Lopez and Elizabeth Bohuski (©DGC, Indiana University 2007). Isolation included an on-column digestion with RNase-free DNase (Qiagen, Hilden, Germany). The cDNA library, consisting of eight pooled specimens, was constructed by GenXPro GmbH (Frankfurt am Main, Germany) using the NEBNext® Ultra™ Directional RNA Library Prep Kit for Illumina® (New England Biolabs). Paired-end sequencing (Illumina NextSeq 500) produced 121'412'590 raw reads (2 × 150 bp). Cleaning of raw reads, i.e., removal of duplicate reads (software fastuniq; Xu et al. 2012), low quality sequence-bases and sequencing artefacts (software cutadapt; Martin 2011), left 108'011'306 high-quality reads (47% GC; Table 1). All filtered reads were *de novo* assembled using the Trinity RNA-Seq (version trinityrnaseq_r20140717, seqType fq, JM 50G, CPU 40, min_contig_length 200, SS_lib_type FR, group_pairs_distance 200, inchworm_cpu 40, bflyHeapSpaceMax 50G; Grabherr et al. 2011). TransDecoder v2.0.1 (Haas et al. 2013) was used to filter the contigs derived from Trinity. A minimum coverage of 50 reads was set for contigs, resulting in a total of 17,759 potential transcripts. Contig length ranged from 201 to 24,010 bp, with a N50 of 2226 bp (Table 1).

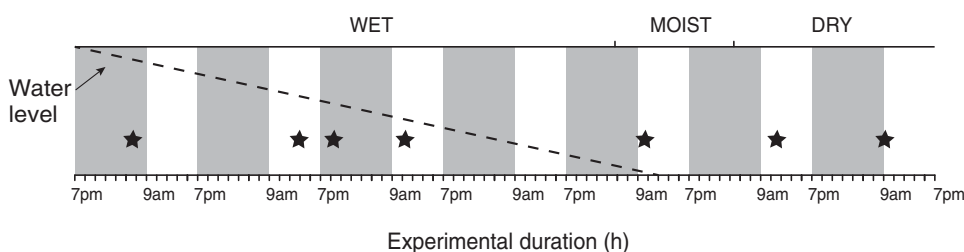


Figure 1. Experimental timeline indicating sampling occasions (stars) at irregular intervals covering various daytimes (grey/white as night/day) and hydrological conditions (wet, moist and dry).

Table 1. Summary of Illumina RNA-Seq transcriptome sequencing, assembly and annotation of larval *Micropterna lateralis* (Stephens, 1837).

Sequencing	Raw reads (paired-end)	121'412'590
	High-quality reads	108'011'306
	GC content (%)	47
Assembly	Assembled contigs	17'759
	Average contig length	1552
	Median contig length	1136
	N50 (bp)	2226
Annotation	Contigs with blast hits	16'848
	Contigs with GO terms	5576
	Contigs with enzyme IDs	1488
	KEGG pathways	110

Functional annotation

The assembled contigs were subjected to similarity search by BLAST (BLAST 2.2.25+; Altschul, Gish, Miller, Myers, and Lipman 1990; Camacho et al. 2009) using the SwissProt and TrEMBL ‘Insecta’ protein sequences from Uniprot (<http://www.uniprot.org/>) followed by ‘Insecta’ mRNA sequences from NCBI for contigs with no match in the Uniprot data-base. BLASTX, and for the latter case BLASTN, were used with an *E*-value cut-off < 1e⁻⁵. Following this procedure, a total of 16,848 contigs had significant matches to Uniprot and NCBI ‘Insecta’ protein sequences (Table 1). We considered these for the transcriptome of *M. lateralis* presented here. Contigs were further assigned gene ontology (GO) terms and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways in Blast2GO v2.5.0

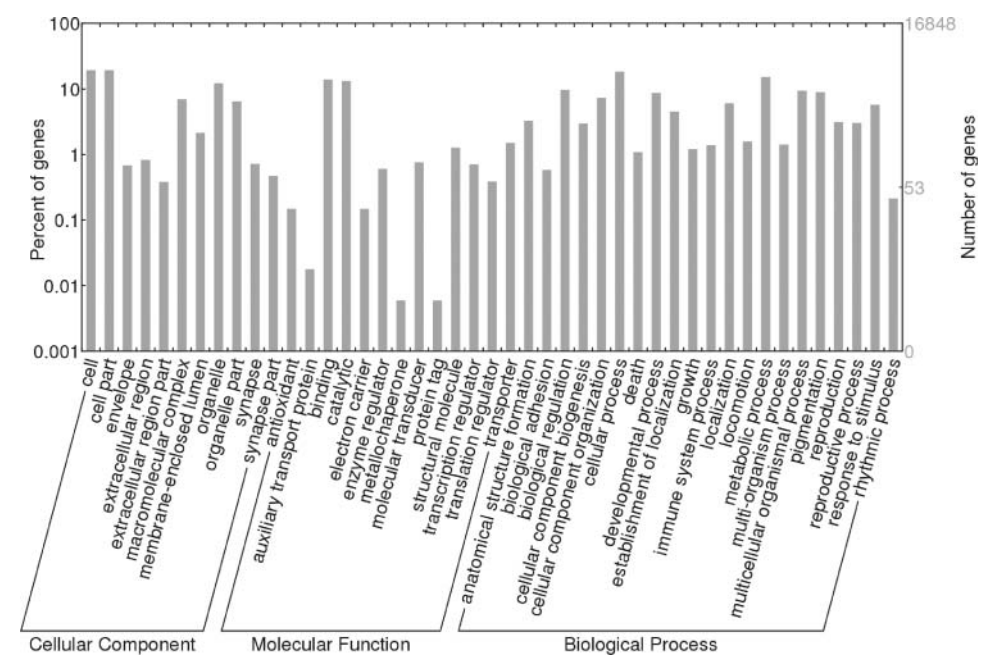


Figure 2. Gene ontology (GO) annotation of *Micropterna lateralis* (Stephens, 1837) transcripts classified into molecular functions, biological processes and cellular components and illustrated using a BGI WEGO plot (Ye et al. 2006).

(Conesa et al. 2005). Figure 2 illustrates the distribution of the 5576 transcripts (33%) that were successfully mapped to at least one GO term for biological processes, molecular functions and cellular components. The most represented terms for biological processes were cellular processes (GO: 0009987, 3090 transcripts) and metabolic processes (GO: 0008152, 2583 transcripts), the most represented terms for molecular functions were binding (GO: 0005488, 2347 transcripts) and catalytic activity (GO: 0003824, 2224 transcripts). Of the 5576 genes, about one-third were assigned Enzyme IDs belonging to 110 different KEGG pathways (Table 1): 682 transcripts were categorised into purine and thiamine metabolism, followed by pathways for biosynthesis of antibiotics (104 transcripts), aminobenzoate degradation (70 transcripts) and T cell receptor signalling (41 transcripts).

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Disclosure statement

No potential conflict of interest was reported by the authors.

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Supplemental material

This Transcriptome Shotgun Assembly project has been deposited at DDBJ/EMBL/GenBank under the accession GELV00000000. The version described in this paper is the first version, GELV01000000. Raw sequence data have been deposited in the Sequence Read Archive (SRA) under the accession number SRP073144. Sequence annotation information is available through MetaCat, the Senckenberg repository.

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